Goal

In this assignment I implemented the hough transform to do line detection on a set of test images

Algorithm

Edge Detector

The Canny edge detector from the previous MP was used to convert the image into Edges. This is stored in util.py

Parameterise

Using a configurable quantisation, a rho, theta parameterised space is created out of the x-y image.

Significant Intersections (Parameter Space)

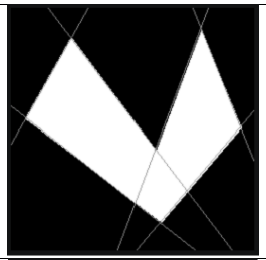
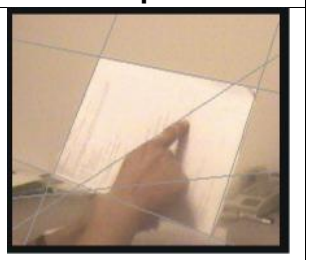
I chose to do thresholding followed by non-maxima surpression to identify the significant intersection points (lines) in the image. This was still in the parameterised space. The configurable values were the size of the non-maxima surpression and the threshold percentage of the interesections that are kept (top 50%, 60% ...). In the end, an array of rho/theta was returned

Unparameterise

The array was converted back into lines and overlayed on the original image for testing and analysis of results. I found 2 points sufficiently far so that the line seems to cross the entire image

Results

If you don't tune the other parameters (threshold, dilation size), then a lower quantisation gives significantly worse results. For quantisation = 128 (128 bins). I begin losing out on lines in even the test cases

	Test	Test2	Input
Quant = 256			
Quant = 128	